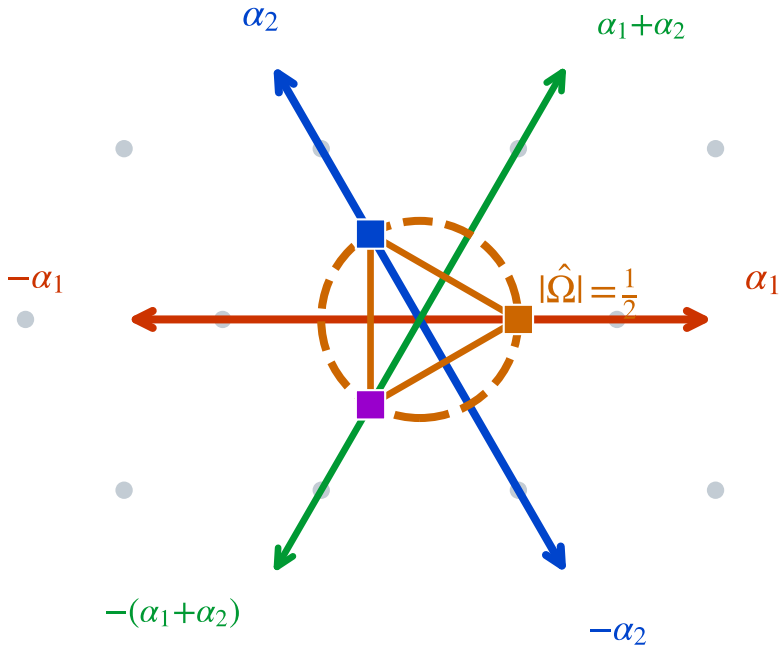


Isometry ↔ algebra (bulk--boundary) via φ : $\mathbb{Z}[\omega]$ (algebra: $120^\circ, S_3$) ↔ $\mathbb{Z}[i]$ (isometry: $\tau=i$) --- $V:H_{\text{bulk}}\rightarrow H_\partial, \quad V^\dagger V=I, |\hat{\Omega}|=1/2$ **preserved**



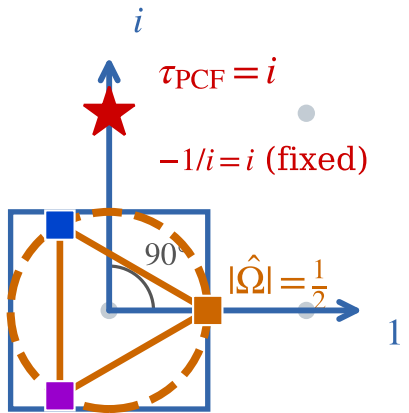
Eisenstein $\mathbb{Z}[\omega]$ (algebra / boundary)
120°, S_3 symmetry
 A_2 root lattice = SU(3)

φ -mediation

$S_3 \rightarrow \mathbb{Z}_4$

bulk ↔ boundary: $V^\dagger V=I$

$\tau_{\text{PCF}}=i$ *fixed*



Gauss $\mathbb{Z}[i]$ (isometry / bulk)
90°, $\mathbb{Z}_4$ symmetry
 $\tau_{\text{PCF}}=i, \ell=1$